5. option(c)

$$Rate = \left(\frac{-1}{2}\right)\frac{d[NH_3]}{dt} = \frac{d[N_2]}{dt} = \left(\frac{1}{3}\right)\frac{d[H_2]}{dt}$$

$$\left(\frac{1}{2}\right)k_1[NH_3] = k_2[NH_3] = \left(\frac{1}{3}\right)k_3[NH_3]$$

$$\left(\frac{3}{2}\right)k_1 = 3k_2 = k_3$$

$$1.5\ k_1 = 3k_2 = k_3$$

6. option(a)

Given :

At low pressure the reaction follows first order, therefore

$$Rate\ \alpha[\text{reactant}]^1$$

$$Rate\ \alpha\ (\text{surface area})$$

At high pressure due to the complete coverage of surface area, the reaction follows zero order.

$$Rate\ \alpha[\text{reactant}]^0$$

Therefore the rate is independent of surface area.

7. option(b)

$$\text{rate} = k[A]^n$$

$$\text{rate} = \frac{-d[A]}{dt}$$

$$\text{unit of rate} = \frac{\text{mol L}^{-1}}{s} = \text{mol L}^{-1}s^{-1}$$

$$\text{unit of rate constant} = \frac{(\text{mol L}^{-1}s^{-1})}{(\text{mol L}^{-1})^n} = \text{mol}^{1-n}\ \text{L}^{n-1}s^{-1}$$

in this case

$$\text{rate} = k[\text{Acetone}]^{3/2}$$

$$n = 3/2$$

$$\text{mol}^{1-(3/2)}\ \text{L}^{(3/2)-1}s^{-1}$$

$$\text{mol}^{-(1/2)}\ \text{L}^{(1/2)}s^{-1}$$

8. option(b)

A catalyst provides a new path to the reaction with low activation energy. i.e., it lowers the activation energy.

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